

Perturbed Assignments and Subgoal-Labeled AI Literacy Instruction in Undergraduate Discrete Mathematics

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Abstract

Generative AI chatbots answer standard coursework accurately enough that a submitted solution no longer shows whether a student reasoned through a problem or delegated it. This ongoing classroom study in undergraduate discrete mathematics ($N = 8$) observes student-chatbot interaction twice and places instruction between the two observations. Two assignments were built with adversarial assignment perturbation, an instructor-led, iterative process that modifies standard questions until a target GenAI chatbot scores below a rubric threshold across three attempts. Between the assignments, one class session taught three subgoal-labeled procedures for diagnosing a hallucinated proof, strengthening a prompt, and collaborating with a chatbot to improve learning gains. Chat logs, pre- and post-AI literacy surveys, and two cued office hours sessions were collected; the logs are being coded with the DRIVE framework for further analysis. In office hours sessions to discuss their AI use with the professor, both consenting students described attempting problems before asking the chatbot to confirm their work and recalled the taught procedures in general terms. The study contributes a second classroom implementation of the perturbation method with a public tool, subgoal-labeled instruction for appraising chatbot-generated proofs, and insights into how undergraduate students use generative AI chatbots for learning to solve proofs.

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1 Problem and Motivation

Generative AI (GenAI) chatbots can pass undergraduate computing coursework: in a theory of computing course, ChatGPT passed the exams while often making “nonsensical claims” in proofs [3]. A correct submission, therefore, says little about how a student used a chatbot, or about whether they learned in the process. AI literacy includes the competencies to “critically evaluate AI technologies” and “communicate and collaborate effectively with AI” [7]. A review of 124 AI literacy studies found only two critical-literacy

interventions for AI tools, of which neither was deployed in an educational setting, leading to an overall lack of empirical intervention studies in post-secondary settings [4]. Instructors of proof-based courses thus need coursework that makes chatbot use observable and classroom-tested instruction on evaluating chatbot output. This study asks: **(RQ1)** How do students’ chatbot interactions on perturbed assignments [2] differ before and after subgoal-labeled AI literacy instruction? **(RQ2)** How do students describe their use of the taught procedures?

2 Background and Related Work

Salim et al. perturbed introductory programming problems to impede LLM-assisted cheating [10]. Adversarial assignment perturbation [2], the author’s prior work, applies the same lever to observation: the chatbot is permitted, its failure is induced on purpose, and the student’s response is recorded. First presented at the SIGCSE TS 2026 Doctoral Consortium, the initial study was executed in a Summer 2025 discrete mathematics course [2]. Eight students’ chat logs were coded with DRIVE, a taxonomy of student-GenAI interaction [9], and independent component analysis yielded two patterns, cognitive scaffolding and dependence, that were associated with grades on the perturbed assignment [2]. Subgoal labels name the function of each step in a worked example and have been studied in computing for programming procedures for increasing lesson content retention [8].

3 Approach and Uniqueness

Setting. The study ran in one section of undergraduate discrete mathematics in Summer 2026, taught by the author, with IRB approval (ID 29198); 8 of 21 students consented (Figure 1).

Perturbed assignments. Following the iterative perturbation process [2], the instructor posed each question to a context-cleared chatbot instance and graded the response with a predetermined rubric. A score of 60% or higher prompted one of three perturbations (Figure 1) and a new test; a question is resistant once the chatbot scores below 60% on three attempts. Lab 3 Part 1 reuses the perturbed counting, induction, and predicate logic questions of [2], validated against Gemini 3.5 Flash (available to students under a university license). Part 2 covered the rest of the course content, and together Part 1 and Part 2 were constructed to serve as a final exam study guide. After the course, the process was implemented as the Perturbation Workbench,¹ a public browser tool that records a verifiable manifest of each run, produced for reproducibility of the iterative perturbation process.

Instruction. One class session between the parts taught three procedures whose steps are subgoal labels, repeated verbatim in a



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¹<https://sgilson7.github.io/perturbation-workbench/>

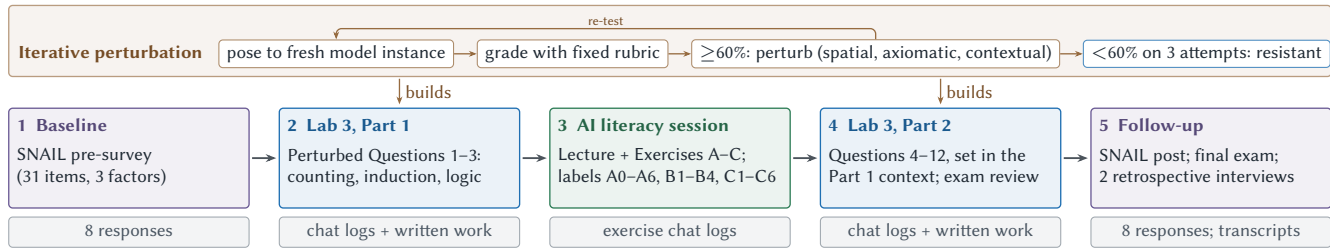


Figure 1: Study design for the 8 consenting students (of 21). The iterative perturbation process [2] builds both assignments.

worked example and the exercises: a hallucination autopsy (A0–A6, such as *test a small case*), four parts of a strong prompt (B1–B4), and a collaboration procedure (C1–C6) in which each chatbot step is interrogated and verified by hand.

Data and analysis. Data comprises chat logs from Part 1, the exercises, and Part 2; written work; the SNAIL AI literacy scale [6], complete before and after for all 8 students; and office-hours interviews with two students after the final exam (60 to 85 minutes each), adapting cued retrospective reporting [11] with each student’s own log and submission as the cue. Each student query receives DRIVE codes, assigned via first and second-round coding techniques. For RQ1, the share of turns that question or correct the chatbot, and the turns that match a taught label, are compared between students in Part 1 and Part 2, following [2]. RQ2 draws on the interviews. Turns will also be tagged with knowledge components [1, 5] from a knowledge graph of the counting unit to locate the concepts students route through the chatbot.

Uniqueness. Earlier perturbation work aimed to impede chatbot use [10], and the first study of perturbed assignments [2] observed use without instruction. This study places instruction between two perturbed assignments [2], compares each student with themselves, and moves subgoal labels from procedures a learner performs [8] to the appraisal of a chatbot’s solution.

4 Results and Contributions

First and second round coding techniques are currently being applied to the entirety of the dataset, to be completed by early November. We present the insights from two tagged office hours sessions (P1, P2).

Comparisons against worked examples P2 said, “I’ll try and solve it myself, and then kind of use it as a confirmation.” P1 said, “I think I’ve got it ... Let me just compare it to the AI so I can be 100% certain.” Both describe written work that preceded the prompt, which a chat log does not record. In the 2025 study, the same confirmatory phrasing came from high-scoring students testing already-written proofs [2].

Procedures recalled in general terms. P1 said, “Did I apply them explicitly? No, but they were in my head ... Is it hallucinating, or ... is my prompt strong enough?” P2 remembered that the session “laid out steps” and could recall none specifically, and described correcting the chatbot as “stuff that I haven’t necessarily thought as much about before.” Neither student named a label, which suggests scoring procedure execution in the logs separately from label recall.

Reflection on reliance. P2 reported getting “in my head a lot about whether or not I’m using AI correctly,” and the session “felt like confirmation” that they were. P1 said that relying on it for certainty “hurt me in other areas” and decided to reduce use: “it’s hard to learn things if you can’t make mistakes.”

Limitations. The interviewees were not selected to span the range of chatbot use. The study covers one summer section, with a single coder to date.

Contributions and next steps. The study contributes (1) a second classroom execution of perturbed assignments [2], against a newer model, (2) a publicly available tool for reproducing the iterative perturbation process, (3) subgoal-labeled AI literacy instruction for proof-based coursework, and (4) insights for curriculum design with regards to generative AI. Coding all eight students’ logs will test whether these accounts hold, and a planned between-subjects study to compare labeled and unlabeled versions of the instruction.

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